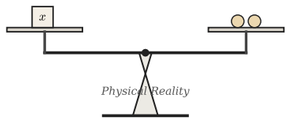
The Epistemic Engine: The Four-Stage Grounded Abstraction Framework

Mathematics is neither arbitrary symbol manipulation nor concrete trivia. It is the contemplation of immutable order abstracted from real being. Human intellects ascend from the sensory referent (*the phantasm*), through the authentic disciplinary name (*nomen*), bound by an operational definition (*the affixing bridge*), into the universal invariant (*species intelligibilis*).

The Realist Foundation

I. The Sensory Referent

The Phantasm: Physical & Spatial Encounter



Nothing in the intellect without first in the senses.

II. The Disciplinary Name

The Formal Signifier: Academic Language

Real Mathematical Terms

- **Equilibrium** = **Vector**
- **Inverse Operation** = **Reciprocal**

Authentic Vocabulary (No Slang)

Vygotsky's scientific concept meets lived reality.

III. The Affixing Bridge

The Functional Tie: Word Tethered to Thing

Operational Affixing Formula

[Real Term] <═══> [Physical Action]

"Inverse Operation" = Action that undoes what was done
"Absolute Value" = Physical ruler distance to zero

Semantic anchoring: active operational definition.

IV. The Abstract Invariant

Species Intelligibilis: Universal Order

The Immutable Laws of Order

$$A = B \Leftrightarrow A \circ C = B \circ C$$

$$a + (-a) = 0 \quad | \quad a(b + c) = ab + ac$$

Grasp of unchangeable truth in the Logos.

I. THE PHILOSOPHICAL FOUNDATION

1. Thomistic Epistemic Realism
Nihil est in intellectu quod non sit prius in sensu (Nothing is in the intellect that was not first in the senses). All human knowledge begins in sensible reality. The human intellect does not possess innate ideas; it abstracts intelligible form from sensory experience.

2. The Role of the Phantasm & *Conversio ad Phantasmata*
St. Thomas Aquinas affirms that the mind never thinks without an internal sensory image (*phantasma*). Even when contemplating the highest abstractions, the intellect must constantly return to the phantasm (*conversio ad phantasmata*, *ST* I, q. 84, a. 7). Teaching detached symbols without sensory referents violates human nature as a composite of body and soul.

3. The Second Degree of Abstraction (Mathematics)
In the Thomistic division of knowledge, mathematics abstracts from sensible qualities (color, heat, weight) while strictly retaining **intelligible matter** (*materia intelligibilis*)—extension, dimension, continuity, and quantity. Grounding mathematical operations in spatial movement, balance beams, and area dissection places mathematics in its true metaphysical degree of abstraction.

4. The Mathematical River (*Logos*)
Mathematics is not a human game invented by social convention; it is an objective reality discovered by the intellect. Creation is intelligible because it originates in the Divine *Logos*. The student enters an eternal stream flowing from antiquity.

II. PSYCHOLINGUISTICS & COGNITIVE SCIENCE

1. Vygotsky's Convergence of Concepts
In *Thought and Language*, Lev Vygotsky proved that authentic intellectual mastery occurs only where the upward development of the **spontaneous (everyday) concept** meets the downward development of the **scientific (academic) concept**.

2. Bruner's EIS Hierarchy
Cognition progresses from **Enactive** (action-based) → **Iconic** (image-based) → **Symbolic** (code-based). Symbolic mastery requires ongoing coordination across all three modes.

3. Embodied Cognition & Perceptual Symbols
Modern neuroscience (Barsalou; Lakoff & Núñez) demonstrates that the brain represents abstract mathematical relations through neural sensorimotor simulations of physical schemas (Source-Path-Goal, Container, Balance).

III. PEDAGOGICAL ARCHITECTURE: THE BALANCED HABITS OF MIND

To prevent any single cognitive movement from hijacking the curriculum, the program dynamically balances Mark Driscoll's complete suite of mathematical habits of mind alongside the NRC's five proficiency strands:

Driscoll's Algebraic Habits of Mind:

- **Doing and Undoing**: Reversing processes, backward deduction, unwrapping operations. Governs 20–25% of decads (equations, factoring, inverse functions, calculus).
- **Building Rules to Represent Functions**: Pattern seeking, covariation, generating input-output rules across tables, graphs, and symbols. Governs rates, linearity, and exponential growth.
- **Abstracting from Computation**: Operating on operations, structural properties (distributive, commutative, associative laws), and algebraic equivalence.

Driscoll's Geometric Habits & NRC Strands:

- **Reasoning with Relationships**: Spatial invariants, symmetry, congruence, and similarity.
- **Balancing Exploration and Deduction**: Synthetic Euclidean proof and axiomatic justification.
- **The NRC Five Strands (*Adding It Up*)**: Integrating Conceptual Understanding, Procedural Fluency, Strategic Competence, Adaptive Reasoning, and Productive Disposition.

IV. THE BRAIDED RIVER ARCHITECTURE: CUMULATIVE RETRIEVAL

Rejecting the commercial model of "beads on a string" (siloed chapters quickly forgotten), the curriculum implements a permanent three-tier braided spiral:

1. **Layer 1: The Daily Micro-Spiral (Problems 4 & 5)**: Every broadsheet plate reserves Problem 4 as *The Bridge* (fusing today's skill with a prior unit) and Problem 5 as *The Cold Retrieval* (active backward synthesis from 20–60 days prior).
2. **Layer 2: Decad Macro-Braiding (Days 9 & 10)**: Day 9 synthesizes the unit with earlier tools in non-routine challenges; Day 10 provides axiomatic retrospective proof.
3. **Layer 3: The 7-Year Vertical Helix (Grades 6–12)**: Foundational tools are never retired; they become the operational substrate of advanced study (e.g., Grade 7 linear balance becomes Grade 8 system equilibrium, Grade 9 quadratic completing the square, Grade 11 logarithmic inversion, and Grade 12 the Fundamental Theorem of Calculus).

V. THE 7-YEAR CURRICULAR SCOPE ARC

Grade 6	Ratio, unit rates, fractional scaling, area shearing invariance, signed coordinate reflection, and statistical centers.
Grade 7	Linear balance (*al − jabr* and *al − muqabala*), signed vectors, distributive property, multi-step systems, and similarity.
Grade 8	Linear functions ($y = mx + b$), slope triangles, simultaneous systems, power laws, roots as inverses, and Pythagorean invariants.
Grade 9 (Alg I)	Function mappings, area dissection factoring, completing the square, quadratic forms, and exponential modeling.
Grade 10 (Geom)	Axiomatic foundations, compass constructions, triangle congruence, similarity, trigonometry, and Cavalieri's principle.
Grade 11 (Alg II)	Polynomial roots, rational asymptotes, partial fractions, logarithmic inverses, unit circle radians, and series.
Grade 12 (Calc)	Function composition, polar/vectors, limits, derivatives, optimization, Riemann accumulation, and the **Fundamental Theorem**.

VI. THE PRODUCTION CODE

1. Tufte Information Design

- *Maximize Data-Ink*: Erase all non-data ink, cartoon mascots, and faux-3D borders.
- *Macro/Micro Reading*: Entire daily territory visible on one plate with zero page turns.
- *Spatial Small Multiples*: Side-by-side state transformations on paper replace temporal video.

2. Hard-Coded Ergonomic Rules

- **Unboxed Workspaces**: Bounding borders removed from sketching areas; delicate, unbordered 5 mm dot grids float freely.
- **Unlined Equation Boxes**: Open, unlined framed fields ($\geq$ 0.45 in tall) with subtle corner labels.
- **Definition Rhythm**: Fixed 6 pt entry, 8 pt inter-definition, and 8 pt exit spacing.
- **Thinking Field**: Entire lower right flank (2.6 in $\times$ 3.0 in) reserved as an open dot-grid canvas.

3. Two-Gate Authenticity Rule (Problem 5)

- **Gate 1 (Authentic Practice)**: Real situations where the unknown is genuinely unknown beforehand (aviation lapse rates, surveying, accounting). Banish pseudorealism.
- **Gate 2 (Pure Contemplation)**: Honest mathematical inquiry (geometric invariants, number theory). Must provide **scaffolded support** (schematics, guided steps) so abstraction is never unassisted.

4. Diagram Architecture Rules

- **Separation of Concerns**: Diagrams depict physical/geometric structures only (max 8-char labels, no procedural prose in figures).
- **1:1 Aspect Ratio**: Always enforce `ax.set_aspect('equal')` and `keepaspectratio` in LaTeX.

5. Certified Print Specifications

- **Trim**: 17 in $\times$ 11 in landscape · **Bleed**: 0.000 in.
- **Gripper Margins**: Left/Right $\geq$ 0.50 in, Top $\geq$ 0.47 in, Bottom $\geq$ 0.85 in.

Immutable Standard for all 70 Decads · 700 Plates